SADIN · SAUDI DIGITAL INNOVATION

Energy Intelligence

An internal reporting tool that reads the company's electricity meters directly from the data warehouse, and the record of how every number on it was proven correct.

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Summary

What this is, in one paragraph

SADIN operates 31 electricity meters across 21 zones. Those meters have produced **109 million** individual measurements, which our data warehouse refines into **4.5 million** clean readings. Until now the only way to look at that data was a vendor-supplied web page that showed very little of it, and a Power BI report that answered management questions but could not be explored freely.

This project delivered a third thing: a private web dashboard, running on our own machine against our own database, that exposes the full detail — every meter, every phase, every individual reading — while reconciling **exactly** to the figures the Power BI report already publishes. Not approximately. To the decimal.

That reconciliation is the point of the whole exercise, and most of the effort went into proving it rather than into building the screens.

2,020,697.5	4,535,918	31	8	0
KWH RECONCILED TO POWER BI	METER READINGS AVAILABLE	METERS 21 ZONES	DASHBOARD PAGES	UNEXPLAINED DISCREPANCIES

01 · Background

Why we built it

The starting point was a tool nobody could get answers out of.

The existing vendor dashboard showed a handful of summary figures and no way to reach the data underneath them. It could not filter by zone, could not show an individual meter's behaviour, and could not display the electrical measurements — current, voltage, power factor — that tell you whether equipment is actually healthy. When a number looked wrong there was no way to check it.

Meanwhile the data warehouse already held everything anyone could want. The gap was purely one of access.

The brief was therefore specific: build something genuinely interactive, covering all 31 meters, showing consumption *and* electrical quality, driven by real database queries rather than pre-computed summaries — and make its numbers agree with the Power BI report that management already trusts.

02 · The product

What the dashboard does

Eight pages, each answering a different operational question. Every page respects the same filters: date range, zone, meter, and time of day.

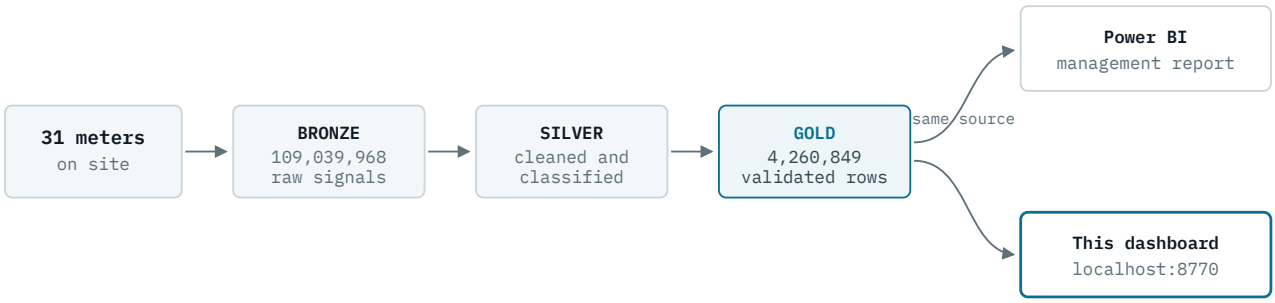
PAGE	QUESTION IT ANSWERS
Command	How much energy did we use, where did it go, and is anything wrong right now?
Energy & Demand	Consumption over time, and power draw at the raw reading cadence — total and per phase.
Bill Check	What the grid invoiced against what our meters actually recorded.
Current	Load on each of the three phases, and how evenly it is shared.
Voltage & Frequency	Is supply quality inside the acceptable band?
Power Factor	Are we drawing power efficiently, and where are we not?
Readings	The individual meter readings themselves, searchable and exportable.
Data & Status	How complete and trustworthy is the underlying data?

Two things distinguish it from the tool it replaces. First, **nothing is pre-aggregated** — every chart is a live query, so any date range and any combination of meters can be examined, not just the ones someone anticipated. Second, **every figure names its own source**: each page header states which database view it reads, and each summary card states which column it sums.

03 · Architecture

How it works

A small program on our own machine, reading our own database. No copies of the data exist anywhere else.



Both reporting tools read the identical gold-layer table. That is *why* their numbers agree — it is structural, not a coincidence to be maintained.

The three pieces

The database. Our existing PostgreSQL warehouse, unchanged. The project added nothing to it and altered nothing in it.

The server (`serve.py`, 935 lines). A small Python program that accepts requests from the browser, turns them into database queries, and returns the results. It opens the database in **enforced read-only mode** and refuses to start if that setting does not take effect — so the dashboard is incapable of altering or deleting company data, by construction rather than by policy.

The interface (`dashboard_updated.html`, 1,589 lines). One self-contained web page. Charts are drawn with Chart.js, a standard open-source library.

> WHY THIS MATTERS COMMERCIALLY
>
> There is no cloud service, no subscription, no third-party processor, and no copy of SADIN's consumption data outside our own network. The dashboard is a file and a small program; running costs are zero and the data never leaves the building.

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04 · The central decision

One source of truth

The single most consequential choice in the project, and the mistake that led to it.

The first version of the dashboard calculated consumption itself — taking each meter's cumulative register, subtracting consecutive readings, and applying sensible rules about gaps and implausible jumps. It produced **1,780,590 kWh**.

The Power BI report said **2,020,697.5 kWh**.

Both were defensible calculations. Only one was the company's number. The warehouse's gold layer already encodes SADIN's own definition of valid consumption — which readings to trust, how to treat gaps, which meters to exclude — in a column called `validated_kwh_consumption`. Power BI simply adds that column up.

The dashboard was rewritten to do exactly the same thing. It does not compute consumption; it reads the company's answer. The two reports now agree because they are summing the identical column of the identical table.

VERIFIED **Dashboard total equals Power BI total**

```
SELECT sum(validated_kwh_consumption)
FROM   gold.energy_consumption_validated;

  2020697.5      <-- of which NORMAL 1,666,455.0 + LONG_GAP 354,242.5
```

The same query the Power BI measure runs. Any future divergence between the two reports would mean the warehouse changed, not that one tool drifted.

The same principle was applied to every threshold on the dashboard. What counts as an acceptable voltage, a poor power factor, or an abnormal current is **not** a textbook value — it is read from SADIN's own `classify_*` functions in the warehouse (19 of them). An early draft used the industry-standard 0.85 power-factor penalty point; the company's own definition is 0.80, and the dashboard was corrected to match. The dashboard states our standards, not generic ones.

05 · Assurance

How the numbers were proven

Agreement between two reports is weak evidence if both read the same table. So the data was traced further back, to the meters themselves.

Verification was done at three independent levels, and a figure was only accepted when all three agreed.

LEVEL	WHAT WAS CHECKED	RESULT
Report vs report	Every one of the 52 Power BI measures re-computed against the database	ALL RECONCILE
Gold vs bronze	Dashboard figures re-derived from the 109M raw signals, bypassing the warehouse entirely	IDENTICAL
Row by row	Every individual reading matched between layers, not just the totals	0 MISMATCHES

The second level is the one worth understanding. Rather than trust that the warehouse transformed the meter signals correctly, the raw signals were pivoted independently — reading the meter's own channel codes for current, voltage, power and power factor — and the dashboard's figures recomputed from scratch. For a sample meter over a five-day window:

VERIFIED **Gold layer faithfully represents the raw meter signals**

```
readings compared        1,579
in raw but not in gold       0
in gold but not in raw       0
values that disagree         0
channels per reading         3   (4,737 = 1,579 x 3, none missing)
```

Repeated for current, voltage, frequency, power, power factor and consumption. Every measurement class passed.

Physical cross-checks

Two checks used physics rather than arithmetic, because they cannot be satisfied by an error that is merely self-consistent:

VERIFIED **Voltages are internally coherent**

```
measured line-to-line / phase-to-neutral   1.731836
square root of 3                           1.732051
                                  error    0.012 %
```

In a correctly wired three-phase system these two voltage families must differ by exactly √3. They do. A transcription or scaling fault anywhere in the chain would break this.

VERIFIED **Consumption matches the meter's own mechanical counter**

```
register at window start   81,127.6 kWh
register at window end     81,838.4 kWh
difference                    710.8 kWh
validated consumption         710.8 kWh   <-- exact
```

The meter's cumulative counter is independent of every calculation in the warehouse. That it moves by precisely the amount the dashboard reports is the strongest single check performed.

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06 · History

How it was built

Delivered across several working sessions. Each round began with a review of the previous build and a list of corrections.

STAGE 1

Discovery and first build

Mapped the warehouse, recovered the definitions of every relevant table and view, and built a first working dashboard covering all 31 meters.

STAGE 2

The reconciliation failure — and the rewrite

Review found the consumption figure disagreed with Power BI. Root cause: the dashboard was calculating consumption rather than reading the company's own validated column. Rewritten around the gold layer. Exact agreement achieved.

STAGE 3

Depth and detail

Added per-reading views, per-phase breakdowns for current, voltage and power factor, and the data-quality page. Thresholds switched from textbook values to SADIN's own classifiers.

STAGE 4

Performance

Slow pages rebuilt. Query times fell from around 90 seconds to under 2. Details in section 08.

STAGE 5

Interface redesign

Shafey produced a redesigned interface; the data layer was reconnected to it and re-verified end to end to confirm no query or figure had changed in the move.

STAGE 6

Independent validation

The bronze-layer and physics checks in section 05 were carried out, meter by meter and measurement by measurement, at Shafey's request.

STAGE 7

Per-phase power and sign-off

Active and reactive power added per phase, matching the treatment already given to current and voltage. Final validation pass completed.

07 · Corrections

Faults found and fixed

Recorded in full, including faults in the dashboard itself. A validation record that lists no failures is not a validation record.

FAULT	EFFECT	RESOLUTION
Consumption re-derived instead of read	Understated total by 240,107 kWh (11.9%)	Rewritten to sum the warehouse's validated column
Power-factor threshold set to 0.85	Used an industry value instead of SADIN's	Read from `classify_power_factor_v2` (0.80)
Status categories hard-coded	79,429 flagged readings hidden from view	Categories now discovered from the data at startup
Reading frequency assumed	Coverage figures exceeded 100%	Each meter's true cadence measured (244 s median)
Wrong imbalance formula	Understated phase imbalance	Replaced with the NEMA MG-1 / IEEE 141 definition
Negative power factor counted on the average channel only	Showed 0 where 587 readings had a phase reversed	Now counts a reading if *any* phase is negative
Chart tooltip read the wrong axis	Showed the row number instead of the value	Corrected; axis labels restored
Error logger crashed on missing files	Dropped browser connections	Type coerced before comparison

A fault found in Power BI, not in the dashboard `RESOLVED BY SHAFEY`

Monthly totals differed slightly between the two reports — December 2024 read 27,401.9 kWh in Power BI against 27,207.4 on the dashboard. Investigation showed Power BI was grouping months by **UTC** while every other part of the system — including Power BI's own weekday and hourly logic — used **Saudi time**. Consumption was being assigned to the wrong month across each month boundary.

Annual totals were unaffected (both summed to 2,020,697.5) and no energy was lost or double-counted; it was purely a boundary misallocation. Shafey corrected the Power BI report to Saudi time, and the two now agree month by month. The change also repaired the *Average Daily Consumption* measure, which had been dividing a UTC-month total by a count of Saudi days.

`08 · Performance`

Making it fast enough to use

The first build was correct but unusably slow. Four changes fixed it, none of which altered a single figure.

PROBLEM	CHANGE	EFFECT
Meter and zone details looked up for all 4.5M rows	Loaded once when the server starts	`90 s → 4 s`
Date filter written so the database index could not be used	Rewritten to compare the timestamp directly	`103 ms → 28 ms`
Some pages read the data 16–26 times over	Rewritten as a single pass	`7.0 s → 1.6 s`
Requests handled one at a time	Server made multi-threaded	`panels load together`

All fourteen data endpoints now respond in well under a second for a typical selection.

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09 · Findings

What the dashboard revealed

Operational issues that were present in the data all along but not visible in any existing report. These are engineering observations, not yet investigated on site.

Load is badly unbalanced across phases `SITE-WIDE`

A three-phase supply should carry roughly equal current on each phase. The industry working limit is around 2%, and motor manufacturers derate above 1%. Across our estate, **more than twenty meters sit between 40% and 70%**, with several above 70%. At one board, phase 1 carries 3.8× the current of phase 3.

Sustained imbalance wastes energy, overheats cabling and shortens equipment life. It is usually correctable by redistributing circuits across phases — typically inexpensive work.

Power factor is the estate's only real quality problem `33–39% OF READINGS`

Voltage, frequency and cumulative energy are flawless — **100%** of readings acceptable. Current is near-perfect. Every quality flag in the system traces back to power factor, which falls below our 0.80 standard on roughly a third of readings.

The pattern is diagnostic: power factor is worst overnight (0.78 at 03:00–06:00, when load is ~4 kW) and best at midday (0.90 at 12:00–13:00, when load is ~9 kW). That is the signature of equipment left energised out of hours with little useful work to do. It suggests scheduling or switched correction rather than a fixed capacitor installation.

One meter's reactive-power channel cannot be trusted DATA INTEGRITY

At meter ZN.15-PB.2 the reported reactive power is inconsistent with that same meter's voltage, current, active power and power factor — all four of which agree with each other. The reported value is roughly 15× too large and of the opposite sign to what the other channels imply.

This is a fault at the meter or in its configuration, not in the warehouse: the raw signal and the processed value are identical, so the pipeline is faithfully carrying through what the device sent. **Recommendation:** confirm whether other meters show the same pattern before any analysis relies on reactive power.

Some measurements were never collected FOR AWARENESS

The apparent-power channels (kVA family) and the meters' own pre-averaged voltage and current channels contain no data at all — the devices do not transmit them. Separately, `reactive_energy_cumulative_kvarh` is present but reads exactly zero on more than half its rows.

The dashboard reports transmission and usefulness as two separate columns for exactly this reason: a channel can be 100% "received" and still carry no information. This also confirms that the Power BI model's omission of those columns was correct rather than an oversight.

10 · Limits

What this does not do

- **It runs on one machine.** The dashboard is started by double-clicking a file and is available at `localhost:8770` on that computer only. Making it available to colleagues would require hosting it on an internal server — straightforward, but not yet done.
- **It shows data up to 2026-07-11.** It always reflects whatever the warehouse currently holds; it does not itself refresh the warehouse.
- **Zone 13 is absent from consumption figures.** Its two meters are excluded from the validated layer by an existing business rule, so the dashboard covers 29 meters for consumption and all 31 for electrical measurements. This asymmetry is deliberate and is what makes the totals match Power BI.
- **Billing analysis is limited.** The warehouse holds one billing period, and the supplier's cost field is unreliable — 18 of 28 invoices imply roughly 0.0002 SAR/kWh against a plausible 0.25. The page reports the kWh comparison and flags the cost data rather than presenting it as fact.
- **The time-of-day filter does not apply to every page.** It filters the individual-readings view but not the summary cards. Known, documented, and fixable on request.

11 · Operation

Running it

1. Open the `LIVE_DASHBOARD` folder.
2. Double-click `START_DASHBOARD.bat`. A black window opens — that is the server; leave it running.
3. The browser opens automatically at `http://localhost:8770` after a few seconds.

4 To stop, close the black window.

The first start takes a few seconds while the server counts rows and loads meter details. Opening the HTML file directly by double-clicking it will *not* work — the page needs the server running to reach the database.

Appendix

Technical reference

Files delivered

FILE	SIZE	PURPOSE
`serve.py`	935 lines	Read-only database server, 14 data endpoints
`dashboard_updated.html`	1,589 lines	The dashboard interface
`START_DASHBOARD.bat`	—	One-click launcher
`README.md`	269 lines	Operating notes and threshold tables
`AUDIT.md`	417 lines	Full audit trail against Power BI and the database
`dashboard.html`	1,187 lines	Superseded first interface, retained for comparison

Data sources

SOURCE	ROWS	USED FOR
`gold.energy_consumption_validated`	4,260,849	All consumption figures (29 meters, 20 zones)
`gold.energy_raw`	4,535,918	All electrical measurements (31 meters)
`bronze.telemetries`	109,039,968	Independent verification only
`silver.classify_*`	19 functions	Every threshold shown on the dashboard

Conventions

- **Time zone.** All dates and times are Saudi local (UTC+3, no daylight saving), matching the warehouse's own `reading_date_saudi` column. Verified as exactly `(UTC + 3 hours)` across all 4,260,849 rows with zero mismatches.

- **Reading cadence.** Median 244 seconds (~4.07 minutes), measured rather than assumed.
- **Long gaps.** Energy recorded across an abnormally long interval is *included* in the totals, matching the warehouse's deliberate treatment (354,242.5 of the 2,020,697.5 kWh).
- **Stack.** Python 3.12 with psycopg 3, PostgreSQL 18, Chart.js 4.4.1. No paid components, no external services.

Method

The warehouse, the classification logic and the Power BI model this dashboard reconciles against are SADIN's existing work. The dashboard application was specified, directed and reviewed by Shafey and implemented with AI pair-programming assistance (Claude), with every figure independently verified against the database as described in section 05. All verification queries are reproducible and recorded in `AUDIT.md`.

SADIN Energy Intelligence — project record, 14 September 2026.

Figures in this document were read from the production warehouse on the date of writing and are reproducible from the queries recorded in `AUDIT.md`.